

OPTIMAL CONTROL

-LECTURE NOTE

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A Brief History of Optimal Control Theory

Simple Optimizayion Problems

- The shortest path joining two points: a straight line segment.
- Isoperimetric problem (i.e., Dido's problem) to find the plane curve of a given length that encloses the largest possible area : the curve is a circle.

1638 Galileo posed two shape problems;

- The shape of a heavy chain suspended between two points (the catenary).
- The shape of a wire such that a bead sliding along its end-part in minimum time (the brachystochrone).

1662 Fermat found that Light chooses the path in minimum time

1685 Newton was the first to solve the shape of a body with minimum drag. He did not publish the result until 1694.

1696 Johann Bernoulli (the Netherlands, professor of univ. of Basel) challenged the Brachystochrone problem.

6 people; Johann Bernoulli, Jakob Bernoulli, Leibniz, l'Hopital, Tschirnhaus, Newton.

1744 Euler(1707-1783) published the collected works (when he was 13)

"Nothing at all takes places in the universe in which some rule of maximum or minimum does not appear."

Euler formulated the problem in general terms.

; Finding the curve $q(t)$ over the internal $a \leq t \leq b$ with given values $q(a)$, $q(b)$ which minimize

$$J = \int_a^b L(t, q(t), \dot{q}(t)) dt \quad (1)$$

For some given function $L(t, q(t), \dot{q}(t))$,

a necessary condition of optimality for the curve q is

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q} \quad (2)$$

→ solution technique; essential geometric

To compute left side of (2), one first evaluate $\frac{\partial L}{\partial \dot{q}}$ treating $\dot{q}$ as an independent variable (not $\dot{q}(t)$, foundation of time), then plugs in $q(t)$ and $\dot{q}(t)$ for q and $\dot{q}$, and finally differentiates with respect to t .

$$\frac{d}{dt} \left[\frac{\partial L}{\partial u}(q(t), \dot{q}(t), t) \right] = \frac{\partial L}{\partial q}(q(t), \dot{q}(t), t) \quad (2')$$

where $L(q, u, t)$ is a function of $q \in \mathbf{R}^n, u \in \mathbf{R}^n, t \in \mathbf{R}^n$ (i.e. $u \equiv \dot{q}(t)$).

- 1755 Lagrange(1736-1813) described an analytical approach (when he was 19)
 → based on “variations” of the optimal curve and using his
 “undetermined multipliers”
 → which led directly to Euler’s necessary condition Euler adopted this
 approach and rename the subject “the calculus of variations”
 Euler - Lagrange equation based on first - order variations, yields only a
 stationary condition.

- 1786 Legendre(1752-1833) studied the second variation and produced a second
 - order necessary condition of optimality(minimum).

$$\frac{\partial^2 L}{\partial \dot{q}^2}(t, q(t), \dot{q}(t)) \geq 0 \quad (3)$$

Legendre derived it for the scalar case, but it was later extended to the
 vector case by Clebsch(1833-1872).

→ now known as Legendre - Clebsch condition.

$$\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \geq 0, \quad 1 \leq i, j < n \quad ; \text{ the Hessian matrix}$$

<Motivation> The Legendre condition is clearly the second order necessary
 condition for a minimum of a function, but eq(2) does not look at all like the first
 order condition for a minimum of that same function.

=> It is natural to ask whether there might be a way to relate the two conditions.

Is it possible that both can be expressed as necessary conditions for a minimum of one and the same function?

=> The maximum principle.

[(Solution)

Let us look at another way of writing (2)

$$\mathbf{H}(q, u, p, t) \equiv \langle p, u \rangle - L(q, u, t) \quad (\text{i})$$

$$\dot{q}(t) \equiv u$$

$$\text{Define } p(t) \equiv \frac{\partial L}{\partial u}(q(t), \dot{q}(t), t) \quad (\text{ii})$$

$$\frac{\partial \mathbf{H}}{\partial p} = u$$

$$\dot{q}(t) = u = \frac{\partial \mathbf{H}}{\partial p} \Rightarrow \dot{q}(t) = \frac{\partial \mathbf{H}}{\partial p} \quad (\text{iii})$$

$$\text{Eq(2')} \quad \frac{d}{dt} \left[\frac{\partial L}{\partial u} \right] = \frac{\partial L}{\partial q}, \quad \frac{d}{dt} [p] = \frac{\partial L}{\partial q}, \quad \frac{\partial \mathbf{H}}{\partial q} = - \frac{\partial L}{\partial q}$$

$$\frac{dp}{dt} = - \frac{\partial \mathbf{H}}{\partial q} \quad (\text{iv})$$

$$\text{We know } \frac{\partial \mathbf{H}}{\partial u} = p - \frac{\partial L}{\partial u}$$

So, equation (ii) gives

$$\frac{\partial \mathbf{H}}{\partial u} = 0 \quad (\text{v})$$

The system of eqs usually written more concisely as

$$\frac{dq}{dt} = \frac{\partial \mathbf{H}}{\partial p}, \quad \frac{dp}{dt} = - \frac{\partial \mathbf{H}}{\partial q}, \quad \frac{\partial \mathbf{H}}{\partial u} = 0 \quad (\text{vi})$$

is exactly equivalent to (2) under completely general condition if $\mathbf{H}$ is defined as (i).

$\mathbf{H}$ is called control Hamitonian

Since $\mathbf{H} = -L + a \text{ linear function of } u$, eq (3) is completely equivalent to

$$\frac{\partial^2 \mathbf{H}}{\partial u^2} \leq 0 \quad \left(\text{i.e., } \frac{\partial^2 \mathbf{H}}{\partial \dot{q}^2} \leq 0 \right) \quad (\text{vii})$$

$\Rightarrow$ check eq(v) and eq(vii)

$$\frac{\partial \mathbf{H}}{\partial u} = 0, \frac{\partial^2 \mathbf{H}}{\partial u^2} \leq 0$$

$\mathbf{H}$ must have a maximum as a function of u .

End]

Before 1838, William Roman Hamilton(1805-1865) had been reformulating the equations of mechanics as a variational principle (least action).

He introduced the function (now known as “Hamiltonian function”)

$$H(t, p, q) \equiv \langle p, \dot{q} \rangle - L(q, \dot{q}, t) \quad (4)$$

$$\text{where } p = \frac{\partial L}{\partial \dot{q}}(t, q, \dot{q}) \quad (5)$$

treat $\dot{q}$ not as an independent variable, but as a function of t, q, p .

$$\rightarrow \dot{q}(t) = \frac{\partial H}{\partial p}(t, p(t), q(t)), \quad \dot{p}(t) = -\frac{\partial H}{\partial q}(t, p(t), q(t)) \quad (6)$$

if and only if the Euler-Lagrange equation(2) is satisfied.

Eq(6) resembles eq(vi), but is not all the same. The difference is that in Hamilton’s definition, $\dot{q}$ is supposed to be treated not as an independent variable but as a function of q, p, t , defined implicitly by the eq (5).

It is easy to see that, if the map $(q, \dot{q}, t) \rightarrow (q, p, t)$ defined by (5) can be inverted, i.e. if we can solve (5) for $\dot{q}$ as a function of p, q, t , then (6) is equivalent to (vi).

Indeed, it is clear that

$$H(q, p, t) \equiv \mathbf{H}(q, u(q, p, t), t)$$

$$\frac{\partial H}{\partial q} = \frac{\partial \mathbf{H}}{\partial q} + \frac{\partial \mathbf{H}}{\partial u} \frac{\partial u}{\partial q} \quad (6')$$

$$\text{Since } \frac{\partial \mathbf{H}}{\partial u} = 0 \text{ for } u = u(q, p, t) \Rightarrow \frac{\partial H}{\partial q} = \frac{\partial \mathbf{H}}{\partial q}$$

Note: It should be clear that the Hamiltonian reformulation of the Euler-Lagrange equations in terms of the “control Hamiltonian” is at least as natural as the classical one, and perhaps even simpler. Moreover, the control formulation has at least one obvious advantage, namely,

(A1) the control version of the Hamilton equations is equivalent to the Euler-Lagrange system under completely general conditions, whereas the classical version only makes sense when the transformation eq(5) can be inverted, at least locally, to solve for $\dot{q}$ as a function of p, q, t .

Eqs(v) and (vii) suggest something. Clearly, H must have a maximum as a function of u . So we state this as a conjecture.

Conjecture M1 ; Besides eq(vi), an additional necessary condition for optimality is that H , as a function of u , have a maximum at $\dot{q}(t)$ for each t .

1838 Jacobi(1804-1851) showed that it could be more compactly written in terms of what is now known as the Hamilton-Jacobi equation (basis of “dynamic programming” developed by Bellman over 100 years later).

$$\phi_t(t, q) + H(t, \phi_q(t, q), q(t)) = 0 \quad (7)$$

If $\phi(t, q)$ is a twice continuously differentiable solution of this equation, then equation (6) defined a regular optimal trajectory, provided that

(a) the strict Legendre-Clebsch condition is satisfied

$$H_{pp} < 0$$

(b) there are no points conjugate to a along trajectory (the Jacobi condition)

Weierstrauss (1815-1897) considered strong variations (δq small, but no restriction on $\delta \dot{q}$). He considered the problem of minimizing an integral I of the form

$$I = \int_a^b L(q(s), \dot{q}(s)) ds \quad \text{for Lagrangian such that } L(q, \dot{q}) \text{ is positively}$$

homogeneous with respect to the velocity $\dot{q}$ (that is, $L(q, \alpha \dot{q}) = \alpha L(q, \dot{q})$ for all $q, \dot{q}$ all $\alpha (\geq 0)$ and does not depend on time)

He introduced the “excess function”

$$E(q, u, \bar{u}) = L(q, \bar{u}) - \frac{\partial L}{\partial u}(q, u) \cdot \bar{u} \quad (8)$$

Depending on three sets of independent variables q, u and $\bar{u}$.

He then proved his side condition:

For a curve $s \rightarrow q^*(s)$ to be solution of the minimum problem,
The function E has to be $E \geq 0$ when evaluated for $q \rightarrow q^*(s)$,
 $u \rightarrow \dot{q}^*(s)$ and a completely arbitrary $\bar{u}$ (is arbitrary value of u).

Note : Homogeneity property $\Rightarrow L(q, u) = \frac{\partial L}{\partial u}(q, u) \cdot u$

So, the “excess function” equally can be written

$$\begin{aligned} E(q, u, \bar{u}) &= L(q, \bar{u}) - \frac{\partial L}{\partial u}(q, u) \cdot \bar{u} - \left[L(q, u) - \frac{\partial L}{\partial u}(q, u) \cdot u \right] \\ &= [L(q, \bar{u}) - \langle p, \bar{u} \rangle] - [L(q, u) - \langle p, u \rangle] \\ &= -\mathbf{H}(q, \bar{u}, p) - (-\mathbf{H}(q, u, p)) = \mathbf{H}(q, u, p) - \mathbf{H}(q, \bar{u}, p) \end{aligned}$$

He essentially discovered “maximum principle”, however he did it using a obscured procedure and then missed some profound implication of his discovery. Weierstrass’ condition, expressed in terms of the control Hamiltonian, simply says that

(MAX) along an optimal curve $t \rightarrow q(t)$, if we define $p(t)$ via (ii), then for every t , the value $u = \dot{q}(t)$ must maximize the (control) Hamiltonian $\mathbf{H}(q(t), u, p(t), t)$ as a function of u .

(Max) can be simplified considerably. Indeed, the requirement that $p(t)$ be defined via (ii) is now redundant: if $\mathbf{H}(q(t), u, p(t), t)$, regarded as a

function of u , has a maximum at $u = \dot{q}(t)$, then $\frac{\partial H}{\partial u}(q(t), \dot{q}(t), p(t), t) = 0$,

so $p(t)$ has to be given by (ii). Moreover, the vanishing of $\frac{\partial H}{\partial u}(q(t), \dot{q}(t), p(t), t)$ is also one of the conditions of (vi). So we can state

(vi) and (MAX) together.

Then,

(Necessary Condition for Optimality: NCO) If a curve $q(t)$ is a solution of the maximization problem, then there has to exist a function $p(t)$ such that the following three conditions hold for all t

$$\begin{aligned}
\dot{q}(t) &= \frac{\partial \mathbf{H}}{\partial p} (q(t), \dot{q}(t), p(t), t) \\
\dot{p}(t) &= -\frac{\partial \mathbf{H}}{\partial q} (q(t), \dot{q}(t), p(t), t) \\
\mathbf{H}(q(t), \dot{q}(t), p(t), t) &= \max_u \mathbf{H}(q(t), u, p(t), t)
\end{aligned} \tag{9}$$

$\mathbf{H}$ must have a maximum as a function of u .

NCO encapsulates in one single statement the combined power of the Euler-Lagrange necessary conditions and Weierstrauss side condition as well as the Legendre condition.

With control Hamiltonian, Weierstrauss's side condition becomes much simpler. The control Hamiltonian does not require the introduction of an "excess function". One does not need to include a formula specifying how $p(t)$ is defined, since eq (9) does this automatically.

If you compare (NCO) with all the other necessary conditions that we had written earlier, quite amazingly the derivatives with respect to the u variable are gone.

Principles of Mathematical Guessing¹ suggest that the existence of the $\frac{\partial L}{\partial u}$ is not needed. → Then there is no longer any reason to insist that the range of values of u be the whole space; any subset of $\mathbf{R}^n$ would do, since the minimization that occurs in eq(9) makes sense over any set.

Conjecture M2 ;

(NCO) should still be a necessary condition for optimality even for problems where $u = \dot{q}(t)$ is restricted to belong to some subset U of $\mathbf{R}^n$, and $L(q, u, t)$ is not required to be differentiable with respect to u .

Now that we have liberated ourselves from the constraint that L be differentiable with respect to u .

¹ Principles of Mathematical Guessing

: If a statement is proved under some specific restrictions but turns out not to involve these restrictions at all, chances are that the restrictions are not needed and the statement is valid even without them.

u (i.e., $\dot{q}(t)$) can be anything?

→ Instead of letting $\dot{q}$ be u, allow $\dot{q}(t)$ to be even more arbitrary.
(It ought to be possible for u (i.e., $\dot{q}$) to be anything, and NCO will still work.)

→ After applying the Principle of Mathematical Guessing again,
we can write $\dot{q} = f(q, u, t)$
a general function

$$\rightarrow \langle p, u \rangle = \langle p, f(q, u, t) \rangle$$

Conjecture M3 ;

(NCO) should still be a necessary condition for optimality even for problems where q is restricted to satisfy a differential equation $\dot{q} = f(q, u, t)$, with the “control function” $t \rightarrow u(t)$ taking values in some set U and allowed to be a “completely arbitrary” U-valued function of t, and the Hamiltonian H now being defined by

$$H(q, u, p, t) = \langle p, f(q, u, t) \rangle - L(q, u, t) \quad (10)$$

A minor modification is needed to make Conjecture M3 to be the Pontryagin maximum principle.

Consider the restriction of the class of admissible function $\dot{q}(t)$ to a subset of $\mathbf{R}^n$, specially so that they also satisfy the set of equations.

$$g(t, q(t), \dot{q}(t)) = 0, \quad t \in [a, b] \quad (11)$$

⇒ a general set of differential-algebraic equations

⇒ sufficient conditions were imposed to ensure that

$\dot{q} = f(t, q(t))$ satisfying equation (11) : problem of Lagrange

since it was solved by the use of Lagrange multipliers.

⇒ The complete solution set the scene for parameterizing the degrees of freedom implicit in the Lagrange problem by considering constraints of the form

$$\dot{q} = f(t, q(t), u(t)) \quad (12)$$

$\downarrow u(t)$; “controls”, which can be chosen at each $t \in (a, b)$ yielding the optimal control problem.

optimal control problem :

Find $u(t)$ on (a, b) to minimize

$$J = \int_a^b L(t, q(t), \dot{q}(t)) dt$$

Subject to $\dot{q} = f(t, q(t), u(t))$

$$t \in (a, b)$$

$$u(t) \in \Omega \subset R^m$$

$$q(a), q(b) \text{ given}$$

1962 Pontryagin

established the necessary conditions in his famous “maximum principle”

$$\dot{q}(t) = \frac{\partial H}{\partial p} \quad , \quad \dot{p}(t) = -\frac{\partial H}{\partial q} \tag{13}$$

$$H(t, p, q, \dot{q}) = \max_{u \in \Omega} \mathbf{H}(t, p, q, u)$$

$$u^* = \max_u \mathbf{H}(t, p, q, u)$$